

Dexter Fernandes

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<https://github.com/Dexter-Fernandes>

Computer Vision Data Scientist

Computer Vision Engineer with 3+ years of experience delivering low-latency vision solutions for Tier 1 companies, including a construction firm in Texas and a UK oil and gas company. Proven track record of collaborating with multinational clients to deploy AI-driven vision solutions in fast-paced dynamic environments. Experienced in applying state-of-the-art AI research while leading testing, QA, and monitoring processes.

WORK EXPERIENCE

Esbaaar • Muscat, Oman

Jan 2023 - Aug 2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception systems in harsh industrial environments, from data curation and model training through deployment and on-site validation, improving F1 by 15% since project takeover while maintaining deterministic 24/7 operation.
- Orchestrated a sensor-data cleansing pipeline for 3.6 million records using Pandas and Apache Airflow, reducing model rollout time from 3 days to under 24 hours.
- Built multimodal data ingestion, cleansing, training, and validation pipelines using Python, Pandas, Apache Airflow, MLflow, DVC, Optuna, and CVAT, cutting model iteration time by 40% and reducing rollout time from 3 days to under 24 hours.
- Refactored legacy services into Dockerised microservices with Kafka-based messaging, structured logging, health checks, retries, and safe restarts, improving fault isolation, increasing MTBF by 35%, and reducing live video pipeline crashes by 67%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, improving maintainability and reducing defects.
- Documented deployment guardrails, failure modes, and recovery procedures, reducing cognitive load for engineers and strengthening maintenance and reliability during operation.
- Implemented Git and Jenkins-based continuous integration and validation workflows for edge inference modules for 24/7 deployments to improve release consistency across hardware targets.
- Collaborated with cross-functional engineering and client teams to translate performance metrics into actionable deployment and optimisation decisions.
- Integrated AI modules with production systems in collaboration with other teams to deliver fault-tolerant edge deployments.
- Engineered production-grade AI software for real-time perception and safety analytics, delivering deterministic performance across 24/7 oil rig operations.
- Validated deterministic model performance under extreme field conditions at a British Petroleum site, ensuring reliability across 24/7 operations.
- Applied Agile and Scrum methodologies to plan sprints, manage deliverables, and ensure iterative software improvement across multidisciplinary teams.
- Led on-site validation and diagnostics on oil-rig deployments, conducting root-cause analysis of model failures and improving overall system stability under environmental stressors.
- Built synthetic image generation and augmentation procedures with OpenCV and SciPy to improve model robustness under harsh industrial conditions, reducing false positives and supporting long-term operational reliability.
- Presented key findings and metrics to leadership and clients to translate complex performance results into actionable insights for deployment and further optimisation.
- Diagnosed and resolved production failures through on-site analysis and structured validation checklists, reducing crash frequency by 67% in live video pipelines.

- Built a Django-based internal computer vision application with a web dashboard to visualise inference, system health, and performance metrics, enabling faster debugging.
- Designed adversarial training experiments (occlusion, lighting variation, motion artefacts), benchmarking robustness via F1 and failure-rate metrics, achieving a 15% F1-score improvement within 3 months.
- Developed a repeatable optimisation workflow (baseline to FP16 to validation) with defined acceptance thresholds, enabling consistent latency, memory, and stability outcomes across deployments.
- Built benchmarking and profiling harnesses to evaluate open-source models (YOLOv5, TensorFlow Model Zoo) and quantify accuracy, latency, and stability pre and post optimisation for regression detection.
- Diagnosed and resolved production issues in Dockerised microservices, improving service continuity through stronger logging, health checks, safe restarts, and workload isolation.
- Implemented reliability patterns for real-time CV services, including timeouts and retries and Kafka pipelines with message validation to maintain real-time behaviour under load.
- Added operational observability using structured logs and metrics, enabling faster root-cause analysis and documented failure modes and recovery procedures to reduce operational overhead.
- Added operational observability using structured logs and metrics, enabling faster root-cause analysis and documented failure modes and recovery procedures to reduce operational overhead.
- Developed an internal web dashboard in Django to surface inference outputs, system health, and performance metrics, enabling faster debugging and root-cause analysis across deployed systems.
- Led validation, benchmarking, and observability for fielded AI deployments, reducing crash frequency by 67% and documenting failure modes, recovery procedures, and deployment guardrails for engineers operating live systems.
- Applied classical image-processing techniques to detect mud and sand splashes on camera lenses, reducing false positives by 13% and improving system reliability in harsh industrial conditions.
- Benchmarked and optimised edge inference workflows across latency, memory, and stability constraints, defining repeatable FP16 validation paths and acceptance thresholds for reliable deployment across hardware targets.
- Led field validation and root-cause analysis on BP oil-rig deployments, diagnosing failures under environmental stressors and translating performance findings into practical fixes that improved robustness and reduced false positives in live operation.

LivNSense Technologies • Bengaluru, India

May 2020 - Dec 2022

Lead Software Engineer - AI • Full-time

- Led a 4-member team to design, train, and deploy production object detection and tracking models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Built low-latency perception pipelines using OpenVINO, and C++17, increasing throughput from 5 FPS to 15 FPS and delivering deterministic real-time inference on Intel VPU hardware under tight power and memory constraints.
- Deployed quantised vision models to rugged Intel VPU edge devices using OpenVINO, delivering deterministic real-time inference across RGB and IR sensor streams in harsh industrial environments.
- Re-architected the perception pipeline and model optimisation workflow, increasing latency from 5 FPS to 15 FPS, enabling real-time detection of safety-critical events.
- Built automated labelling, dataset versioning, and performance-tracking pipelines, repurposing detection models for annotation and cutting manual labelling effort by 80%, accelerating iteration cycles in modern Python.
- Developed synthetic data generation and augmentation tools to improve edge-case coverage and cut manual labelling time by 60% using OpenCV and Albumentations.
- Conducted systematic quantisation and optimisation experiments benchmarking accuracy–latency–memory trade-offs and reducing model size by 50% while meeting real-time constraints.
- Led a 4-engineer team to design, train, and deploy CNN-based object detection and tracking models for industrial safety applications, reducing false positives by 33% and achieving sub-50 ms inference latency on edge devices.
- Optimised neural network architectures using quantisation-aware training, structured pruning, and layer fusion reducing model size by 50% with Keras and Intel OpenVINO.
- Developed adversarial-condition training and validation conditions simulating occlusion, lighting variation, and motion artefacts, cutting performance variance by 25% and increasing resilience.

- Collaborated with hardware, software, and QA teams to integrate AI modules into production systems, mentoring engineers on optimisation and deployment practices to improve delivery consistency.
- Engineered production-grade edge perception software for safety-critical environments, delivering deterministic real-time behaviour on Intel VPU devices.
- Mentored engineers and collaborated with hardware, software, and QA teams to integrate perception modules into deployed systems and improve delivery consistency.
- Presented key findings and metrics to leadership and clients, translating complex performance results into actionable insights for deployment and further optimisation.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.
- Defined and standardised FP16/INT8 optimisation paths (including QAT, pruning, and layer fusion), reducing model size by 50% while preserving accuracy under tight memory and latency constraints.
- Implemented optical flow-based ego-motion estimation feeding motion representations into CNNs to classify vehicle state in production pipelines.
- Built asynchronous inference pipelines using OpenVINO and C++17 to maximise throughput on Intel VPU edge hardware under real-time latency and power constraints.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.
- Designed data handling and retention practices aligned with GDPR principles, including data minimisation and controlled access, within production computer vision systems.
- Owned the full machine learning pipeline from data ingestion and labelling through training, optimisation, deployment, monitoring, and field validation, accelerating iteration cycles and improving deployment reliability.
- Provided technical leadership to a 4-member engineering team, setting experimentation standards, reviewing model performance, and guiding optimisation strategies for production deployments.
- Implemented an asynchronous inference pipeline in C++ with OpenVINO, overlapping pre-processing, inference, and post-processing to reduce end-to-end latency and improve pipeline utilisation on Intel GPU targets.
- Profiled and optimised C++ inference and pre and post-processing hotspots using Intel VTune Profiler for targeted performance fixes in our CNN inference pipeline.
- Integrated Boost C++ libraries into production pipelines, resolving dependency and build issues and maintaining reproducible deployments.
- Defined optimisation workflows spanning quantisation-aware training, pruning, layer fusion, FP16/INT8 deployment, and profiling, reducing model size by 50% while meeting accuracy, latency, and memory targets.
- Built automated data, labelling, and performance-tracking pipelines, repurposing older models for annotation and cutting manual labelling effort by 80% and also developed synthetic-data tooling to improve edge-case coverage and reduce manual labelling time by 60%.
- Worked closely with hardware, software, and QA teams in a fast-moving start-up environment to integrate perception modules into deployed systems, improve delivery consistency, and guide experimentation and optimisation for production releases.

PROJECTS

Counsel Copilot

Feb 2026 - Present

- Building Counsel Copilot, a Python-based prototype for legal document editing, retrieval, and research, designed to support contract review workflows.
- Implementing a FastAPI-backed service layer with OpenAI models, PageIndex-powered RAG, and python-docx document handling to generate grounded answers, propose structured edits, and preserve DOCX formatting integrity.
- Designing a human-in-the-loop redlining workflow that proposes structured edits, previews document changes, and requires explicit approval before updates are applied.
- Developing a citation-grounded legal research flow that combines query interpretation, document retrieval, and structured answer generation to improve traceability and user trust.

- Creating a DOCX processing pipeline for section-aware parsing, controlled edit generation, and reliable export of revised legal drafts while preserving document structure.
- Targeting retrieval hit rate@5 above 85%, citation support rate above 95%, and faithfulness above 0.90 on a held-out evaluation set built from CUAD - a contract documents dataset.

ROS2 SLAM & Nav2 Simulation

Feb 2026

- Built a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate LiDAR-centric SLAM, LiDAR+RGB fusion SLAM, and autonomous navigation in structured scenarios.
- Benchmarked RTAB-Map, LIO-SAM, Cartographer, and GLIM across 2D and 3D LiDAR configurations, comparing mapping and localisation behaviour to support system selection and iteration.
- Developed ROS2 nodes, launch files, and parameterised configurations in C++ and Python, using tf2 and message synchronisation to maintain frame consistency and debug transform timing issues.
- Implemented point cloud processing with PCL and Open3D for filtering, segmentation, and registration, and added an EKF-based localisation layer with robot_localization to fuse wheel odometry and IMU for smoother odom-to-base tracking during LiDAR-only SLAM.
- Containerised the stack with Docker on Linux and managed versioning in Git to support reproducible builds and consistent execution across development environments.

End-to-End Damage Severity & Claim Assessment

- Developed an end-to-end automotive claims assessment pipeline combining computer vision-based damage detection with a trained severity-classification model, achieving accurate triage for real-world vehicle damage scenarios.
- Built a lightweight repair-insight engine that transforms model outputs into insurer-friendly explanations, using structured reasoning + LLM-generated narratives to deliver clear, interpretable claim summaries.
- Designed the entire workflow (data preprocessing, model training, evaluation, and a user-facing inference module) demonstrating practical understanding of insurance claims processes.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

Sep 2023 - Sep 2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on the Clearpath Husky platform, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios to assess mapping and localisation behaviour in simulation before real-world deployment.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL for filtering, segmentation, and registration.
- Developed and maintained ROS2 nodes using tf2 and message synchronisation, and debugged tf trees and frame connectivity.

Coursework:

- Hands-on work with inverse kinematics and closed-loop control on a laboratory robotic arm platform.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

Jul 2016 - May 2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, perception systems, and robotics-adjacent engineering.

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

Jul 2019

AWARDS & SCHOLARSHIPS

Best Speaker award

ABC Toastmasters club

Jan 2020

SKILLS

- **SLAM & 3 Perception:** Cartographer, Ceres, Eigen, Filtering, GLIM, LiDAR SLAM, LIO-SAM, Point cloud processing, Pose graph optimisation, Registration, RTAB-Map, Scan matching, Segmentation
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, AWS, CI/CD, Docker, DVC (Data Version Control), Intel VPUs, Microsoft Azure, MLFlow
- **Programming & Tooling:** Bash, C++17, CMake, colcon, Django, FastAPI, Git, Intel VTune, Linux, NumPy, Pandas, PG vector, PostgreSQL, Python, Qt, SciPy
- **Deep Learning and Optimisation:** Bayesian Optimisation, Convolutional Neural Networks, Data Augmentation, Layer fusion, MultiRocket, Object Detection & Tracking, ONNX, OpenVINO, Optuna, Post-training Quantisation, PyTorch, Quantisation-Aware Training, Scikit-Learn, Structured pruning, TensorFlow, TensorRT, Vision Transformer (ViT), YOLO
- **Edge AI and Inference:** Intel Core i7 (VTC7252-7C4IP), Intel Movidius Myriad VPU, NVIDIA Jetson AGX Xavier
- **Professional Skills:** Agile framework, Experiment-driven development, Stakeholder communication, Technical leadership
- **Embedded and Low-level:** 8051 Assembly, ADC/DAC interfacing, Arduino C, Embedded C, Interrupt-driven I/O, Timers and counters, UART serial
- **Point Cloud & Vision Processing:** Denoising, Open3D, OpenCV, Point Cloud Library, Scikit-Image
- **Robotics Stack:** Closed-loop control, Closed-loop robot control, Gazebo (Harmonic), Inverse kinematics, Localisation, Nav2, rclcpp, rclpy, Real-time perception systems, ROS2 (Robot Operating System 2), Sensor integration, SLAM, tf2
- **Large Language Models:** Citation-grounded response generation, Error analysis, Experimentation and eval, Information extraction, Model fine-tuning, Retrieval-augmented generation (RAG), Search and Retrieval, Workflow automation
- **Computer Vision & Image Processing:** Data augmentation, Image processing, LiDAR extrinsics calibration, Object detection, Open3D, OpenCV, PCL (Point Cloud Library), Sensor calibration
- **AI-Assisted Development:** ChatGPT Codex, Claude Code, Pompt engineering